FastLoRA: Low-Rank Adaptation with Adaptive Rank Allocation

Maya Iyer, Daniel Okafor and Lin Wei. Fictional paper written for testing PaperMind.

Abstract

We present FastLoRA, a parameter-efficient fine-tuning method that allocates low-rank adapter ranks adaptively across the layers of a language model under a fixed total rank budget. On the NLU-Bench suite of nine language understanding tasks, FastLoRA reaches an average accuracy of 84.3 while training only 1.2 million parameters, which is 0.017% of the 7-billion-parameter base model. Training takes 14 GPU-hours on a single NVIDIA A100 GPU.

1 Introduction

Fine-tuning large language models is expensive because every weight must be updated and stored. Low-rank adaptation (LoRA) reduces this cost by training small rank-decomposition matrices next to frozen weights. However, standard LoRA assigns the same rank to every layer, even though layers differ widely in how much they need to change for a downstream task.

We observe that the lower layers of a transformer require almost no adaptation, while the upper layers and the attention output projections carry most of the task-specific signal. This motivates allocating rank where it matters instead of uniformly. Our contributions are: (1) an adaptive rank allocation rule based on gradient energy, (2) a budgeted training schedule that prunes unused rank during training, and (3) an evaluation on NLU-Bench showing near full fine-tuning accuracy at a tiny fraction of the cost.

2 Related Work

LoRA introduced low-rank updates for efficient fine-tuning. Adapter layers and prefix tuning are earlier parameter-efficient approaches. Some recent work prunes adapter ranks after training, but it still pays the full cost of training at the maximum rank. FastLoRA instead decides the rank during training, so the unused capacity is never paid for.

3 Method

FastLoRA keeps the base model frozen and attaches a low-rank adapter to each of the 32 transformer layers. The total rank budget is fixed at 64, so the average rank per layer is 2. At the start of training every layer receives rank 2. Every 200 steps we measure the gradient energy of each adapter, defined as the squared Frobenius norm of its accumulated gradient, and move one rank unit from the lowest-energy layer to the highest-energy layer.

Rank is reassigned by truncating or extending the adapter matrices with a small random initialization. Because the total budget never changes, the number of trainable parameters stays constant at about 1.2 million throughout training. We use a learning rate of 2e-4 with cosine decay, a batch size of 32, and train for 3 epochs.

4 Experiments

We evaluate on NLU-Bench, a suite of nine tasks including sentiment analysis, natural language

inference, paraphrase detection and question answering. The base model has 7 billion parameters. We compare against standard LoRA with rank 8 (4.2 million trainable parameters) and full fine-tuning (all 7 billion parameters). All methods use the same data, optimizer and number of epochs. Each configuration is run with three random seeds and we report the mean.

5 Results

FastLoRA achieves an average NLU-Bench accuracy of 84.3. Standard LoRA with rank 8 reaches 82.9 while training 4.2 million parameters, so FastLoRA is 1.4 points more accurate with roughly three times fewer trainable parameters. Full fine-tuning reaches 85.0 but updates all 7 billion parameters and needs far more memory. FastLoRA therefore recovers about 96% of the gap between LoRA and full fine-tuning.

The learned rank allocation concentrates in the top eight layers, which receive 41% of the total budget. An ablation with a fixed uniform rank of 2 gives only 81.7, confirming that adaptive allocation, rather than the smaller parameter count, explains the improvement. Training FastLoRA takes 14 GPU-hours on one A100 GPU.

6 Limitations

FastLoRA becomes unstable when the total rank budget is set below 32, where training loss sometimes diverges. We only evaluated English-language tasks and did not test open-ended generation beyond summarization. The gradient energy criterion also adds a small bookkeeping overhead of about 3% per training step.

7 Conclusion

FastLoRA shows that where rank is placed matters as much as how much rank is used. By reallocating a fixed budget toward the layers that need it, it matches most of the accuracy of full fine-tuning at a tiny cost. Future work includes multilingual evaluation and combining adaptive rank with quantized base models.

References

[1] A. Author. Low-rank adaptation of language models. 2021. [2] B. Writer. Parameter-efficient transfer learning. 2019. [3] C. Scholar. The NLU-Bench evaluation suite. 2023.